VAERS COVID-19 Adverse Event Clustering

Purpose

Identify recurring symptom-report patterns in COVID-19 VAERS narratives using unsupervised text clustering, then relate those clusters back to structured clinical and reporting features for interpretation.

Overview

1. **Setup & Imports**
2. **Load Data**
3. **Load Precomputed Comorbidity Indicators**
4. **Symptom Text Preprocessing**
5. **TF-IDF + LSA**
6. **Stability Analysis**
7. **Final Clustering**
8. **Cluster Interpretation**
9. **Cluster Characterization**
10. **Export**

1. Setup & Imports

In [5]:
```
# ============================================================
# 1. SETUP
# ============================================================

import os
import re
import sys
import json
import math
import warnings
from collections import defaultdict

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import TruncatedSVD
from sklearn.preprocessing import Normalizer, OneHotEncoder
from sklearn.cluster import MiniBatchKMeans
from sklearn.metrics import silhouette_score, adjusted_rand_score
```

```
from sklearn.model_selection import StratifiedKFold, GridSearchCV, cross_val
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.metrics import (
    average_precision_score,
    f1_score,
    precision_score,
    recall_score,
    confusion_matrix
)
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassif

import scipy.sparse as sp

warnings.filterwarnings("ignore")

print("Python executable:", sys.executable)
print("Python version:", sys.version.split()[0])
print("Pandas version:", pd.__version__)
```

```
Python executable: /Users/ariellerothman/Desktop/Sperm Cell Analysis/.venv/b
in/python
Python version: 3.12.10
Pandas version: 2.2.3
```

2. Load Data

Read the cleaned VAERS dataset produced by `Data preparation.ipynb`. The notebook will try `Outputs/df_clean_imputed.csv` first, then `Outputs/df_clean_engineered.csv` as fallback.

In [ ]:
```
# 2. LOAD DATA
from pathlib import Path

HERE = Path().resolve()  # project root (the folder containing this notebook
OUTDIR = str(HERE / "Outputs")
os.makedirs(OUTDIR, exist_ok=True)

candidate_paths = [
    str(HERE / "Outputs" / "df_clean_imputed.csv"),
    str(HERE / "Outputs" / "df_clean_engineered.csv"),
]

data_path = next((p for p in candidate_paths if os.path.exists(p)), None)
if data_path is None:
    raise FileNotFoundError(
        "Expected one of these files but none were found:\n" + "\n".join(car
    )

df_clean = pd.read_csv(data_path, low_memory=False)
```

```python
print("Dataset loaded:")
print(f"Path:  {data_path}")
print(f"Shape: {df_clean.shape}")
```

```
Dataset loaded:
Shape: (986096, 42)

Columns: ['VAERS_ID', 'DIED', 'L_THREAT', 'ER_VISIT', 'ER_ED_VISIT', 'HOSPIT
AL', 'DISABLE', 'BIRTH_DEFECT', 'SERIOUS', 'AGE_YRS', 'HOSPDAYS', 'NUMDAYS',
'ONSET_INTERVAL', 'RECVDATE', 'VAX_DATE', 'ONSET_DATE', 'DATEDIED', 'SEX',
'STATE', 'VAX_TYPE', 'SYMPTOM_TEXT', 'LAB_DATA', 'OTHER_MEDS', 'CUR_ILL', 'H
ISTORY', 'PRIOR_VAX', 'ALLERGIES', 'MAX_DOSE', 'DOSE_COUNT', 'MULTI_DOSE',
'UNKNOWN_DOSE', 'MULTI_MANUFACTURER', 'MANU__JANSSEN', 'MANU__MODERNA', 'MAN
U__NOVAVAX', 'MANU__PFIZER\\BIONTECH', 'MANU__UNKNOWN MANUFACTURER', 'YEAR',
'MONTH', 'NUMDAYS_MISSING', 'ONSET_INTERVAL_MISSING', 'AGE_YRS_MISSING']

Data types:
VAERS_ID                        int64
DIED                            int64
L_THREAT                        int64
ER_VISIT                        int64
ER_ED_VISIT                     int64
HOSPITAL                        int64
DISABLE                         int64
BIRTH_DEFECT                    int64
SERIOUS                         int64
AGE_YRS                       float64
HOSPDAYS                      float64
NUMDAYS                       float64
ONSET_INTERVAL                float64
RECVDATE                       object
VAX_DATE                       object
ONSET_DATE                     object
DATEDIED                       object
SEX                            object
STATE                          object
VAX_TYPE                       object
SYMPTOM_TEXT                   object
LAB_DATA                       object
OTHER_MEDS                     object
CUR_ILL                        object
HISTORY                        object
PRIOR_VAX                      object
ALLERGIES                      object
MAX_DOSE                      float64
DOSE_COUNT                      int64
MULTI_DOSE                      int64
UNKNOWN_DOSE                    int64
MULTI_MANUFACTURER              int64
MANU__JANSSEN                   int64
MANU__MODERNA                   int64
MANU__NOVAVAX                   int64
MANU__PFIZER\BIONTECH           int64
MANU__UNKNOWN MANUFACTURER      int64
YEAR                            int64
MONTH                           int64
NUMDAYS_MISSING                 int64
ONSET_INTERVAL_MISSING          int64
AGE_YRS_MISSING                 int64
dtype: object
```

3. Load Comorbidity Indicator Columns

In [ ]:
```
# ============================================================
# 3. LOAD PRECOMPUTED COMORBIDITY INDICATORS
# ============================================================

import os
import pandas as pd

from pathlib import Path
HERE   = Path().resolve()  # project root
OUTDIR = str(HERE / "Outputs")
os.makedirs(OUTDIR, exist_ok=True)

COMORB_PATH = str(HERE / "Outputs" / "comorbidity_indicators.csv")

# Detect currently available indicator columns
existing_indicator_cols = [
    c for c in df_clean.columns
    if "__" in c or c.endswith("_MISSING")
]

print(f"Existing indicator columns already in df_clean: {len(existing_indica

if not os.path.exists(COMORB_PATH):
    raise FileNotFoundError(
        f"Comorbidity indicator file not found: {COMORB_PATH}\n"
        "Run Data preparation notebook first to generate comorbidity_indicat
    )

comorb_df = pd.read_csv(COMORB_PATH, low_memory=False)

if "VAERS_ID" not in comorb_df.columns:
    raise ValueError("comorbidity_indicators.csv must contain VAERS_ID for m

indicator_cols = [
    c for c in comorb_df.columns
    if c != "VAERS_ID" and ("__" in c or c.endswith("_MISSING"))
]

missing_in_df = [c for c in indicator_cols if c not in df_clean.columns]

if not missing_in_df:
    print("All comorbidity indicator columns are already present. Skipping m
else:
    merge_cols = ["VAERS_ID"] + missing_in_df

    # Use one row per VAERS_ID to prevent accidental row multiplication
    comorb_merge = comorb_df[merge_cols].drop_duplicates(subset=["VAERS_ID"]

    df_clean = df_clean.merge(
        comorb_merge,
        on="VAERS_ID",
        how="left",
```

```
        validate="m:1"
    )

    # Fill unmatched IDs as 0 and cast to compact integer dtype
    for c in missing_in_df:
        df_clean[c] = df_clean[c].fillna(0).astype("int8")

    print(f"Merged {len(missing_in_df)} indicator columns from comorbidity_i

final_indicator_cols = [
    c for c in df_clean.columns
    if "__" in c or c.endswith("_MISSING")
]

print(f"Total indicator columns now in df_clean: {len(final_indicator_cols)}

print("Sample indicator columns:")print(final_indicator_cols[:20])
```

```
Saved CSV: /Users/ariellerothman/Desktop/Capstone/Outputs/comorbidity_indica
tors.csv shape: (986096, 63)
Parquet not saved (missing engine). CSV is saved and sufficient.
Parquet error: ImportError("Unable to find a usable engine; tried using: 'py
arrow', 'fastparquet'.\nA suitable version of pyarrow or fastparquet is requ
ired for parquet support.\nTrying to import the above resulted in these erro
rs:\n - Missing optional dependency 'pyarrow'. pyarrow is required for parqu
et support. Use pip or conda to install pyarrow.\n - Missing optional depend
ency 'fastparquet'. fastparquet is required for parquet support. Use pip or
conda to install fastparquet.")

Indicator count: 62

Top indicators by prevalence:
                                  count  percent
PRIOR_VAX_MISSING                940654   95.392
CUR_ILL_MISSING                  862779   87.494
ALLERGIES_MISSING                743721   75.421
LAB_DATA_MISSING                 726769   73.702
OTHER_MEDS_MISSING               589519   59.783
HISTORY_MISSING                  585378   59.363
MANU__PFIZER\BIONTECH            459918   46.640
MANU__MODERNA                    452935   45.932
OTHER_MEDS__other                273957   27.782
LAB_DATA__other                  195633   19.839
HISTORY__other                   173203   17.565
ALLERGIES__other                 143636   14.566
AGE_YRS_MISSING                  102852   10.430
ALLERGIES__drug_allergy           77750    7.885
MANU__JANSSEN                     73515    7.455
CUR_ILL__other                    71218    7.222
HISTORY__cardiovascular           63285    6.418
HISTORY__respiratory              59886    6.073
HISTORY__metabolic                56111    5.690
LAB_DATA__vitals_procedures       53601    5.436
OTHER_MEDS__statin                43514    4.413
OTHER_MEDS__thyroid_meds          41952    4.254
HISTORY__endocrine_thyroid        37831    3.836
HISTORY__musculoskeletal          36235    3.675
HISTORY__mental_health            35486    3.599
HISTORY__neurologic               27809    2.820
OTHER_MEDS__psych_meds            26052    2.642
HISTORY__gastrointestinal         25701    2.606
OTHER_MEDS__diabetes_meds         24974    2.533
HISTORY__cancer_immunocompromised 18421    1.868

HISTORY OTHER examples (n=10):
0     Medical History/Concurrent Conditions: No adve...
2                                                Anemic
9                                       unknown || no
10                            acne, astigmatism, myopia
16                                     none documented
18                                  borderline diabetic
19    low blood sugar, low blood pressure || Comment...
20                                Iron deficiency anemia
25     Environmental allergies, low back herniated disc
```

```
33                                      Type 2 Diabetic
Name: HISTORY, dtype: object

CUR_ILL OTHER examples (n=10):
1                        Post nasal drip, but no illness
9                                                 No || no
19     none || Low blood pressure (Chronic or long-st...
32                                   Seasonal allergies
43                                         No illnesses
50                 COVID19+ 3 weeks prior to the vaccine
54     Fever and chills, but was not diagnosed with a...
74     Nausea in the morning prior to coming to work ...
101                                         None reported
133    Taking antibiotics for bladder infeciton  at t...
Name: CUR_ILL, dtype: object

ALLERGIES OTHER examples (n=10):
2                                  Xylocaine, Novacaine
7                                                Biaxin
9                                              No || no
11    Gabapentin Adhesive Duloxetine || Gabapentin, ...
12                                          Hydrocodone
15                                            cucumbers
18    Allergies: Morphine Spinal and Lumigan (Bimato...
21                                               Iodine
22        Seasonal allergies, cats, dogs, dust, ragweed
25                             Dust, mold, grass, trees
Name: ALLERGIES, dtype: object

PRIOR_VAX OTHER examples (n=10):
8      vaso-vagal type symptoms, similar presentation...
66                             syncopal episode as child
109                         Injection Site pain/swelling
148                                   Pneumonia vaccine
221    She gets the same adverse events every month w...
222                                          arm soreness
235                                 allergy shot in 2016
303            Tenderness at sight, headache, body aches
310          TDaP, age 20 swollen arm at injection site.
431             fever, diarrhea, sore throat, runny nose
Name: PRIOR_VAX, dtype: object

LAB_DATA OTHER examples (n=10):
5     She took Benadryl 50 mg po.  No medical tests ...
6     Basic Chemistry, EKG, and vitals were all norm...
8                                    Trransferred to ED
9               Unknown but sent to SICU for monitoring
11    I received IV Benadryl, steroids, Tylenol, Zof...
28                        Sent for COVID 19 PCR testing
44                                             (-) || no
53              blood drawn.  un sure of  test ordered.
64                                   12/16/20-EKG WNL
67    none || Test Date: 20201216; Test Name: pulse;...
Name: LAB_DATA, dtype: object

OTHER_MEDS OTHER examples (n=10):
```

```
1     Multivitamin and vitamin D supplements || VITA...
2     B Stress complex, Vit. D, Folic Acid, Krill oi...
3                                                  latex
7     Lisinopril, HCTZ, lipitor, amyitrypline, lexap...
9                                  unknown || protonix
10    Benzoyl peroxide 10% gel, Biofreeze topical ge...
11    DULOXETINE (CYMBALTA) 60 MG CAPSULE    Take 1 ...
13    Fish oils (2 caps); beet root powder; multivit...
18    Dorzolamide HCL/Timolol Maleate Ophthalmic  So...
19    Nuvaring || Vit D, Women's Multivitamin, Nuvar...
Name: OTHER_MEDS, dtype: object
```

4. Symptom Text Preprocessing (Clustering-Optimized)

Clean and normalize `SYMPTOM_TEXT` for clustering. Two important differences from severity prediction preprocessing:

- **Administrative terms are removed** these terms appear in nearly every record and would dominate the TF-IDF feature space without contributing clinical signal
- **Outcome terms are kept** unlike severity prediction where these cause label leakage, here they may meaningfully separate phenotype clusters (e.g., cardiac arrest cluster vs. mild injection-site reaction cluster)

Records with fewer than 3 tokens after cleaning are excluded from clustering as near-empty narratives carry no useful signal.

In [ ]:

```python
# ============================================================
# 4. SYMPTOM TEXT PREPROCESSING - CLUSTERING OPTIMIZED
# ============================================================
# Produces two text columns:
# - SYMPTOM_TEXT_CLEAN: base cleaning only
# - SYMPTOM_TEXT_CLEAN_CLUSTER: base cleaning + boilerplate removal
#
# Key difference from severity prediction:
# Outcome terms (hospitalization, death, ICU, serious) are intentionally
# kept here. In severity prediction they cause label leakage; for
# clustering, they help define clinically meaningful phenotype groups.

import re
import pandas as pd

# ---------------------------------------------------
# Base text cleaning
# ---------------------------------------------------
_url_re = re.compile(r"http\S+|www\S+|https\S+")
_email_re = re.compile(r"\S+@\S+")
_num_re = re.compile(r"\d+")
_keep_re = re.compile(r"[^a-z0-9\s\-]")
_ws2 = re.compile(r"\s+")

def clean_symptom_text(x):
```

```
    """
    Base-level cleaning for symptom text.
    Steps: lowercase, remove URLs/emails, replace digits with __NUM__,
    remove non-alphanumeric except hyphens, collapse whitespace.
    """
    if pd.isna(x) or x == "":
        return ""
    x = str(x).strip().lower()
    x = _url_re.sub(" ", x)
    x = _email_re.sub(" ", x)
    x = _num_re.sub(" __NUM__ ", x)
    x = _keep_re.sub(" ", x)
    x = _ws2.sub(" ", x).strip()
    return x

df_clean["SYMPTOM_TEXT_CLEAN"] = df_clean["SYMPTOM_TEXT"].apply(clean_symptc

# ----------------------------------------------------
# Clustering-specific administrative cleanup ONLY
# (No supervised leakage scrubbing)
#
# Removes: vaccine boilerplate, batch/lot info, report templates,
#          dosing language, and generic metadata terms that appear
#          across nearly all VAERS records and carry no discriminative
#          signal for grouping clinical phenotypes.
# Keeps:   all clinical/symptom terms, including outcome-related ones.
# ----------------------------------------------------
admin_phrases = [
    # vaccine / manufacturer boilerplate
    r"\bcovid vaccine\b",
    r"\bcovid immunisation\b",
    r"\bmrna moderna\b",
    r"\bmoderna covid\b",
    r"\bpfizer[-\s]?biontech\b",
    r"\bbiontech\b",
    r"\bmoderna\b",
    r"\bpfizer\b",
    r"\bjanssen\b",
    r"\bnovavax\b",
    r"\bbnt\b",
    r"\bmrna\b",
    r"\bvaccine\b",
    r"\bcovid\b",

    # lot / batch / product fields
    r"\blot number\b",
    r"\bbatch number\b",
    r"\bbatch lot\b",
    r"\bvaccine lot\b",
    r"\blot\b",
    r"\bbatch\b",
    r"\bdosage form\b",
    r"\bform\b",
    r"\broute administration\b",
    r"\broute admin\b",
    r"\bunspecified route\b",
```

```
    r"\badministration\b",
    r"\bintramuscular\b",
    r"\bsuspension injection\b",
    r"\bsuspension\b",

    # generic case-report boilerplate
    r"\bpatient received\b",
    r"\bpatient experienced\b",
    r"\bsubject experienced\b",
    r"\bspontaneous report\b",
    r"\bmedical history\b",
    r"\bmedical information\b",
    r"\bfollowing information\b",
    r"\bdescribes occurrence\b",
    r"\boccurrence\b",
    r"\bprovided\b",
    r"\bcontactable\b",
    r"\bcase\b",
    r"\breporter\b",
    r"\breported\b",
    r"\breport\b",
    r"\boutcome\b",
    r"\bunknown date\b",
    r"\bdate patient\b",
    r"\bdate\b",

    # dosing / process boilerplate
    r"\bsingle dose\b",
    r"\bdose\b",
    r"\bbooster\b",
    r"\badmin\b",
    r"\bprophylactic vaccination\b",
    r"\breceived\b",
    r"\bgiven\b",
    r"\badministered\b",
    r"\buse\b",

    # recurring vague metadata terms
    r"\bunknown\b",
    r"\binformation\b",
    r"\bdescribed\b",
    r"\bdescribes\b",
    r"\boccurred\b",
    r"\bpatient\b",
    r"\bsubject\b",

    # concomitant / medication boilerplate
    # "concomitant medications" is standard VAERS report phrasing and appear
    # across all record types — it signals nothing clinical for clustering.
    r"\bconcomitant\b",
    r"\bmedication\b",
    r"\bmedications\b",
]

admin_regexes = [re.compile(p) for p in admin_phrases]
```

```python
def remove_admin_terms(text):
    """
    Remove vaccine-brand boilerplate and administrative phrasing for cluster

    Targets terms so common across all VAERS records (vaccine names, 'patier
    received', 'dose', 'concomitant', etc.) that they carry no discriminativ
    signal for clustering. Unlike supervised leakage scrubbing, outcome term
    are kept intentionally.

    Args:
        text: Cleaned symptom text string.

    Returns:
        Text with administrative/boilerplate terms removed.
    """
    if pd.isna(text) or text == "":
        return ""
    text = str(text).lower()
    for rgx in admin_regexes:
        text = rgx.sub(" ", text)
    text = _ws2.sub(" ", text).strip()
    return text

df_clean["SYMPTOM_TEXT_CLEAN_CLUSTER"] = (
    df_clean["SYMPTOM_TEXT_CLEAN"]
    .apply(remove_admin_terms)
)

# ----------------------------------------------------
# Token count after cluster cleanup
# ----------------------------------------------------
df_clean["CLUSTER_TOKEN_COUNT"] = (
    df_clean["SYMPTOM_TEXT_CLEAN_CLUSTER"]
    .str.split()
    .str.len()
    .fillna(0)
    .astype(int)
)

# Clustering-specific subset (rows with >= 3 tokens)
df_cluster = df_clean[df_clean["CLUSTER_TOKEN_COUNT"] >= 3].copy()

# ----------------------------------------------------
# Summary
# ----------------------------------------------------
print("=" * 60)
print("SYMPTOM TEXT PREPROCESSING - CLUSTERING OPTIMIZED")
print("=" * 60)
print(f"\nRows in full dataset:         {len(df_clean):,}")
print(f"Rows retained for clustering: {len(df_cluster):,}")
print(f"Rows dropped (sparse):        {len(df_clean) - len(df_cluster):,}")
print(f"\nBoilerplate REMOVED:")
print("  ✗ Vaccine brand names (Pfizer, Moderna, Janssen, etc.)")
print("  ✗ Lot/batch numbers, dosing/route language")
print("  ✗ Case-report template phrasing (patient received, administered, et
print("  ✗ Concomitant / medication boilerplate")
```

```
print(f"\nClinical signal KEPT:")
print("  ✓ Symptoms (fever, headache, chest pain, etc.)")
print("  ✓ Outcome terms (serious, hospital, death, ICU)")
print("  ✓ Comorbidity mentions (diabetes, asthma, etc.)")
print("=" * 60)

# ----------------------------------------------------
# Save cleaned text variants
# ----------------------------------------------------
df_clean[
    [
        "VAERS_ID",
        "SYMPTOM_TEXT",
        "SYMPTOM_TEXT_CLEAN",
        "SYMPTOM_TEXT_CLEAN_CLUSTER",
        "CLUSTER_TOKEN_COUNT"
    ]
].to_csv(
    f"{OUTDIR}/symptom_text_clean_variants_cluster_only.csv",
    index=False
)
print(f"\nSaved: symptom_text_clean_variants_cluster_only.csv")
print(f"Text fields available: {[c for c in df_clean.columns if 'SYMPTOM' in
```

```
============================================================
STEP 17: SYMPTOM TEXT PREPROCESSING - CLUSTERING OPTIMIZED
============================================================

Rows in full dataset:        986,096
Rows retained for clustering: 955,355
Rows dropped (sparse):        30,741

Preprocessing removed:

Preprocessing KEPT (for pure symptom clustering):
  ✓ Clinical symptoms (fever, headache, chest pain, etc.)
  ✓ Outcome terms (serious, hospital, death, ICU)
  ✓ Comorbidity mentions (diabetes, asthma, etc.)
============================================================

Saved: symptom_text_clean_variants_cluster_only.csv
Text fields available: ['SYMPTOM_TEXT', 'SYMPTOM_TEXT_CLEAN', 'SYMPTOM_TEXT_
CLEAN_CLUSTER', 'CLUSTER_TOKEN_COUNT']
```

5. TF-IDF Vectorization & Dimensionality Reduction (LSA)

Vectorize `SYMPTOM_TEXT_CLEAN_CLUSTER` using TF-IDF (5,000 features, unigrams + bigrams, sublinear TF scaling, `min_df=10`), fitted only on the clustering-eligible rows. Reduce the sparse TF-IDF matrix to 200 dense LSA dimensions via Truncated SVD, then L2-normalize.

In [ ]:

```
# ============================================================
# 5. TF-IDF + LSA ON CLUSTER-CLEANED TEXT
# ============================================================
# Preserve row-index mapping for safe label assignment later.

cluster_indices = df_cluster.index.values
np.save(f"{OUTDIR}/cluster_row_indices.npy", cluster_indices)

tfidf_cluster = TfidfVectorizer(
    max_features=5000,
    ngram_range=(1, 2),
    min_df=10,
    max_df=0.9,
    stop_words="english",
    sublinear_tf=True,
    norm="l2",
    strip_accents="unicode",
    token_pattern=r"(?u)\b[a-zA-Z][a-zA-Z\-]+\b|__NUM__"
)

# Use df_cluster (rows with >= 3 tokens), not df_clean
X_tfidf_cluster = tfidf_cluster.fit_transform(df_cluster["SYMPTOM_TEXT_CLEAN
print("TF-IDF shape:", X_tfidf_cluster.shape)
print(f"Fit on {len(df_cluster)} rows with >= 3 tokens")

# Save TF-IDF + vocabulary
sp.save_npz(f"{OUTDIR}/symptom_text_tfidf_cluster_cleaned.npz", X_tfidf_clus
pd.DataFrame({"term": tfidf_cluster.get_feature_names_out()}).to_csv(
    f"{OUTDIR}/tfidf_vocabulary_cluster_cleaned.csv", index=False
)

# SVD / LSA
svd_cluster = TruncatedSVD(n_components=200, random_state=42)
X_lsa_cluster = svd_cluster.fit_transform(X_tfidf_cluster)
X_lsa_cluster = Normalizer(copy=False).fit_transform(X_lsa_cluster)

np.save(f"{OUTDIR}/symptom_text_lsa_200d_cluster_cleaned.npy", X_lsa_cluster
np.save(f"{OUTDIR}/svd_components_200d_cluster_cleaned.npy", svd_cluster.com

print(f"X_lsa_cluster shape: {X_lsa_cluster.shape}")
print("Cluster indices saved for alignment")
```

```
TF-IDF shape: (955355, 5000)
Fit on 955355 rows with >= 3 tokens
X_lsa_cluster shape: (955355, 200)
Cluster indices saved for alignment in Step 21
```

6. Cluster Count Selection: Elbow + Silhouette

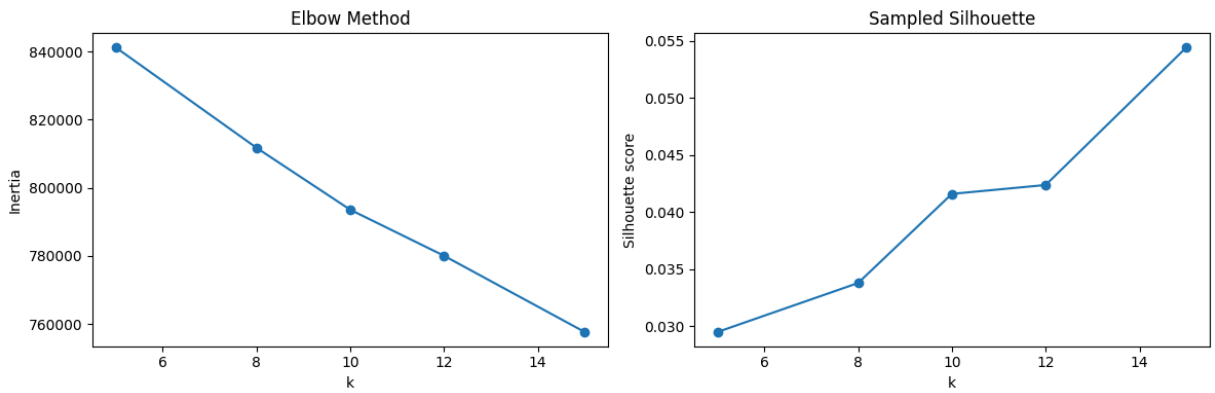
Evaluate k ∈ {5, 8, 10, 12, 15} using two complementary metrics:

- **Inertia (elbow method)**
- **Silhouette score**

Set `best_k` in the stability cell based on these results before proceeding.

In [ ]:
```
# ==============================================================
# 6. CLUSTER COUNT SELECTION: ELBOW + SAMPLED SILHOUETTE
# ==============================================================

rng = np.random.default_rng(42)
silhouette_sample_size = 20000
silhouette_idx = rng.choice(X_lsa_cluster.shape[0], size=silhouette_sample_s
X_sil = X_lsa_cluster[silhouette_idx]

k_values = [5, 8, 10, 12, 15]
elbow_results = []

for k in k_values:
    km = MiniBatchKMeans(
        n_clusters=k,
        random_state=42,
        batch_size=10000,
        n_init=3
    )
    labels_full = km.fit_predict(X_lsa_cluster)
    labels_sil = labels_full[silhouette_idx]
    sil = silhouette_score(X_sil, labels_sil)

    elbow_results.append({
        "k": k,
        "inertia": km.inertia_,
        "silhouette": sil
    })

    print(f"k={k:2d} | silhouette={sil:.4f} | inertia={km.inertia_:.2e}")

elbow_df = pd.DataFrame(elbow_results)
elbow_df.to_csv(f"{OUTDIR}/cluster_selection_metrics_cluster_cleaned.csv", i

fig, ax = plt.subplots(1, 2, figsize=(12, 4))
ax[0].plot(elbow_df["k"], elbow_df["inertia"], marker="o")
ax[0].set_title("Elbow Method")
ax[0].set_xlabel("k")
ax[0].set_ylabel("Inertia")
ax[1].plot(elbow_df["k"], elbow_df["silhouette"], marker="o")
ax[1].set_title("Sampled Silhouette")
ax[1].set_xlabel("k")
ax[1].set_ylabel("Silhouette score")
plt.tight_layout()
plt.show()
```

```
k= 5 | silhouette=0.0295 | inertia=8.41e+05
k= 8 | silhouette=0.0338 | inertia=8.12e+05
k=10 | silhouette=0.0416 | inertia=7.94e+05
k=12 | silhouette=0.0424 | inertia=7.80e+05
k=15 | silhouette=0.0544 | inertia=7.58e+05
```



7. Stability Analysis

Fit K-Means with the chosen `best_k` across 5 different random seeds on a 100,000-row subsample. Compute pairwise Adjusted Rand Index (ARI) to measure how consistently the same cluster structure emerges regardless of initialization.

In [ ]:

```
# ============================================================
# 7. STABILITY ANALYSIS (ARI ACROSS RANDOM SEEDS)
# ============================================================

# Set chosen k here after reviewing elbow + silhouette
best_k = 8

stability_sample_size = min(20000, X_lsa_cluster.shape[0])
stability_idx = rng.choice(X_lsa_cluster.shape[0], size=stability_sample_siz
X_stability = X_lsa_cluster[stability_idx]

seeds = [1, 7, 21, 42, 88]
labels_by_seed = {}

for seed in seeds:
    km = MiniBatchKMeans(
        n_clusters=best_k,
        random_state=seed,
        batch_size=10000,
        n_init=3
    )
    labels_by_seed[seed] = km.fit_predict(X_stability)

stability_results = []
for i, seed_a in enumerate(seeds):
    for seed_b in seeds[i+1:]:
        ari = adjusted_rand_score(labels_by_seed[seed_a], labels_by_seed[see
        stability_results.append({"seed_a": seed_a, "seed_b": seed_b, "ARI":

stability_df = pd.DataFrame(stability_results)
print(stability_df)
print("\nMean ARI:", stability_df["ARI"].mean())

stability_df.to_csv(f"{OUTDIR}/cluster_stability_ari_cluster_cleaned.csv", i
```

```
   seed_a  seed_b       ARI
0       1       7  0.433252
1       1      21  0.457860
2       1      42  0.411774
3       1      88  0.751655
4       7      21  0.495183
5       7      42  0.426743
6       7      88  0.474293
7      21      42  0.459517
8      21      88  0.443869
9      42      88  0.391653

Mean ARI: 0.4745799228929858
```

8. Final Clustering Model

Fit the final K-Means on the full LSA-reduced dataset (`X_lsa_cluster`) using `best_k` and `n_init=5` . Cluster labels are mapped back to `df_clean` using the saved row indices. Rows excluded from clustering (near-empty narratives) receive `NaN` for the `CLUSTER` column.

In [1]:

```
# ============================================================
# 8. FINAL CLUSTERING MODEL ON FULL DATASET
# ============================================================

# Load the saved cluster indices (rows that had >= 3 tokens)
cluster_indices = np.load(f"{OUTDIR}/cluster_row_indices.npy")

final_km = MiniBatchKMeans(
    n_clusters=best_k,
    random_state=42,
    batch_size=10000,
    n_init=5
)

cluster_labels_subset = final_km.fit_predict(X_lsa_cluster)

# Initialize CLUSTER column as NaN in full dataset
df_clean["CLUSTER"] = np.nan

# Map labels back using saved indices
df_clean.loc[cluster_indices, "CLUSTER"] = cluster_labels_subset

print(f"\nClustered rows: {df_clean['CLUSTER'].notna().sum()}")
print(f"Unclustered (sparse): {df_clean['CLUSTER'].isna().sum()}\n")

cluster_sizes = df_clean["CLUSTER"].value_counts().sort_index()
print(cluster_sizes)
(cluster_sizes / df_clean["CLUSTER"].notna().sum()).rename("proportion").to_
    f"{OUTDIR}/cluster_sizes_cluster_cleaned.csv"
)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[1], line 6
      1 # ============================================================
      2 # 8. FINAL CLUSTERING MODEL ON FULL DATASET
      3 # ============================================================
      4
      5 # Load the saved cluster indices (rows that had >= 3 tokens)
----> 6 cluster_indices = np.load(f"{OUTDIR}/cluster_row_indices.npy")
      8 final_km = MiniBatchKMeans(
      9     n_clusters=best_k,
     10     random_state=42,
     11     batch_size=10000,
     12     n_init=5
     13 )
     15 cluster_labels_subset = final_km.fit_predict(X_lsa_cluster)

NameError: name 'np' is not defined
```

9. Cluster Interpretation: Top Terms

Reconstruct approximate TF-IDF centroid weights by projecting K-Means cluster centroids from LSA space back to term space.

In [ ]:

```
# ============================================================
# 9. CLUSTER INTERPRETATION: RECONSTRUCTED TOP TERMS
# ============================================================

feature_names = pd.read_csv(f"{OUTDIR}/tfidf_vocabulary_cluster_cleaned.csv"
svd_components = np.load(f"{OUTDIR}/svd_components_200d_cluster_cleaned.npy"

reconstructed_centroids = final_km.cluster_centers_ @ svd_components

top_terms_rows = []
for cluster_id in range(best_k):
    centroid = reconstructed_centroids[cluster_id]
    top_idx = np.argsort(centroid)[::-1][:20]
    top_terms = feature_names[top_idx]

    print(f"\nCluster {cluster_id} top terms:")
    print(", ".join(top_terms))

    for rank, term in enumerate(top_terms, start=1):
        top_terms_rows.append({"cluster": cluster_id, "rank": rank, "term":

top_terms_df = pd.DataFrame(top_terms_rows)
top_terms_df.to_csv(f"{OUTDIR}/cluster_top_terms_cluster_cleaned.csv", index
```

```
Cluster 0 top terms:
vaccination, spontaneous, action taken, route, action, time, concomitant, sp
ontaneous consumer, taken, treatment, pain, included, consumer, female, year
-old, mar, medications, site, vaccination site, arm

Cluster 1 top terms:
minutes, pt, felt, feeling, called, bp, symptoms, ems, home, benadryl, bette
r, dizzy, stated, started, hr, went, throat, mg, receiving, water

Cluster 2 top terms:
single, unspecified, immunisation, consumer, medications, concomitant medica
tions, concomitant, non, relevant, single immunisation, hcp, included, immun
isation relevant, immunization, attempts, non-serious, follow-up attempts, n
on hcp, consumer non, follow-up

Cluster 3 top terms:
days, pain, left, right, chest, day, shot, started, symptoms, rash, second,
weeks, heart, developed, severe, swelling, nd, week, dizziness, blood

Cluster 4 top terms:
prior, vaccination, prior vaccination, events, weeks, arm, receive, did, did
receive, included, left, tested, diagnosed, vaccination tested, weeks prior,
left arm, year-old, immunisation, vaccines weeks, medications

Cluster 5 top terms:
expired, adverse, error, stored, temperature, storage, vial, product, expire
d product, adverse event, event, freezer, excursion, recommended, temperatur
e excursion, ml, time, expiration, did, doses

Cluster 6 top terms:
fever, headache, injection, chills, site, arm, pain, injection site, aches,
body, hours, day, fatigue, days, body aches, red, nausea, sore, rash, muscle

Cluster 7 top terms:
positive, hospital, admitted, tested positive, acute, ed, pt, tested, home,
cough, symptoms, presented, vaccinated, admission, breakthrough, days, infec
tion, respiratory, pneumonia, admitted hospital
```

10. Cluster Characterization

Profile each cluster across clinical and demographic variables to understand what distinguishes the phenotype groups:

- **Seriousness rate** — proportion of `SERIOUS=1` within each cluster
- **Age** — mean and distributional summary
- **Sex** — proportion by gender
- **Manufacturer** — vaccine brand distribution across clusters
- **Comorbidities** — mean prevalence of each binary comorbidity indicator per cluster

In [ ]:
```
# ============================================================
# 10. CLUSTER CHARACTERIZATION
# ============================================================
```

```python
# Cluster vs serious outcome
cluster_serious = pd.crosstab(df_clean["CLUSTER"], df_clean["SERIOUS"], norm
print("\nCluster vs SERIOUS")
print(cluster_serious)
cluster_serious.to_csv(f"{OUTDIR}/cluster_vs_serious_cluster_cleaned.csv")

# Cluster vs sex
if "SEX" in df_clean.columns:
    cluster_sex = pd.crosstab(df_clean["CLUSTER"], df_clean["SEX"], normaliz
    print("\nCluster vs SEX")
    print(cluster_sex)
    cluster_sex.to_csv(f"{OUTDIR}/cluster_vs_sex_cluster_cleaned.csv")

# Cluster vs age
if "AGE_YRS" in df_clean.columns:
    cluster_age = df_clean.groupby("CLUSTER")["AGE_YRS"].describe()
    print("\nCluster age summary")
    print(cluster_age)
    cluster_age.to_csv(f"{OUTDIR}/cluster_age_summary_cluster_cleaned.csv")

# Cluster vs manufacturer
manufacturer_cols = [c for c in df_clean.columns if c.startswith("MANU__")]
if manufacturer_cols:
    cluster_manu = df_clean.groupby("CLUSTER")[manufacturer_cols].mean()
    print("\nCluster vs manufacturer")
    print(cluster_manu)
    cluster_manu.to_csv(f"{OUTDIR}/cluster_vs_manufacturer_cluster_cleaned.c

# Cluster vs comorbidities
comorb_cols = [c for c in df_clean.columns if "__" in c and not c.startswith
if comorb_cols:
    cluster_comorb = df_clean.groupby("CLUSTER")[comorb_cols].mean()
    print("\nCluster vs comorbidities (first 10 cols)")
    print(cluster_comorb.iloc[:, :10])
    cluster_comorb.to_csv(f"{OUTDIR}/cluster_vs_comorbidities_cluster_cleane

# Cluster summary
cluster_summary = df_clean.groupby("CLUSTER").agg({
    "SERIOUS": "mean",
    "AGE_YRS": "mean",
    "VAERS_ID": "count"
}).rename(columns={"SERIOUS": "SERIOUS_RATE", "AGE_YRS": "MEAN_AGE", "VAERS_

print("\nCluster summary")
print(cluster_summary)
cluster_summary.to_csv(f"{OUTDIR}/cluster_summary_cluster_cleaned.csv")
```

Cluster vs SERIOUS

SERIOUS	0	1
CLUSTER		
0.0	0.922577	0.077423
1.0	0.704846	0.295154
2.0	0.906960	0.093040
3.0	0.763668	0.236332
4.0	0.858460	0.141540
5.0	0.990920	0.009080
6.0	0.938145	0.061855
7.0	0.229411	0.770589

Cluster vs SEX

SEX	F	M	U
CLUSTER			
0.0	0.601956	0.264253	0.133791
1.0	0.683519	0.309516	0.006966
2.0	0.598139	0.291171	0.110689
3.0	0.650773	0.338696	0.010531
4.0	0.736076	0.255161	0.008762
5.0	0.465750	0.416860	0.117390
6.0	0.768461	0.224691	0.006848
7.0	0.506869	0.486591	0.006540

Cluster age summary

	count	mean	std	min	25%	50%	75%	max
CLUSTER								
0.0	122315.0	55.810128	14.837144	0.08	51.0	51.0	66.0	109.0
1.0	101215.0	43.483201	19.664617	0.08	28.0	42.0	58.0	105.0
2.0	79493.0	52.766412	13.241814	0.08	51.0	51.0	58.0	102.0
3.0	248134.0	48.262638	19.348558	0.08	34.0	49.0	63.0	107.0
4.0	41656.0	47.837802	16.428813	0.08	36.0	49.0	60.0	102.0
5.0	85569.0	44.396722	24.007159	0.08	22.0	50.0	64.0	110.0
6.0	196541.0	48.799073	16.471761	0.08	36.0	49.0	61.0	103.0
7.0	80432.0	62.761468	19.632363	0.17	51.0	66.0	77.0	107.0

Cluster vs manufacturer

	MANU__JANSSEN	MANU__MODERNA	MANU__NOVAVAX	MANU__PFIZER\BIONTECH
\				
CLUSTER				
0.0	0.165327	0.822851	0.000270	0.014667
1.0	0.094719	0.371556	0.001245	0.532550
2.0	0.000994	0.005271	0.000226	0.995861
3.0	0.072223	0.424484	0.001161	0.503756
4.0	0.000792	0.003217	0.000024	0.996087
5.0	0.024822	0.568477	0.000666	0.407706
6.0	0.077607	0.583522	0.000692	0.337024
7.0	0.079235	0.398722	0.000584	0.521869

	MANU__UNKNOWN MANUFACTURER
CLUSTER	
0.0	0.005935
1.0	0.001936
2.0	0.061515
3.0	0.002728
4.0	0.000336

```
5.0                        0.001017
6.0                        0.002015
7.0                        0.007248

Cluster vs comorbidities (first 10 cols)
         HISTORY__cardiovascular  HISTORY__respiratory  HISTORY__metabolic
\
CLUSTER
0.0                     0.006884              0.005044            0.004194
1.0                     0.074436              0.079247            0.060841
2.0                     0.035601              0.045815            0.039689
3.0                     0.067830              0.068499            0.062317
4.0                     0.061840              0.081765            0.058191
5.0                     0.017997              0.010670            0.017097
6.0                     0.059178              0.081535            0.061290
7.0                     0.218023              0.113102            0.163915

         HISTORY__endocrine_thyroid  HISTORY__musculoskeletal  \
CLUSTER
0.0                        0.003932                  0.005265
1.0                        0.039500                  0.035509
2.0                        0.032569                  0.029562
3.0                        0.046495                  0.044718
4.0                        0.065969                  0.047676
5.0                        0.004488                  0.005224
6.0                        0.056289                  0.050931
7.0                        0.054804                  0.067946

         HISTORY__mental_health  HISTORY__gastrointestinal  HISTORY__kidney
\
CLUSTER
0.0                     0.003197                   0.002772         0.002502
1.0                     0.051613                   0.028790         0.007222
2.0                     0.023788                   0.013485         0.005070
3.0                     0.041401                   0.029980         0.008125
4.0                     0.047676                   0.026167         0.005353
5.0                     0.008777                   0.006147         0.003634
6.0                     0.045970                   0.032904         0.005098
7.0                     0.064900                   0.065757         0.058820

         HISTORY__clotting  HISTORY__autoimmune
CLUSTER
0.0               0.002322             0.002412
1.0               0.004525             0.010848
2.0               0.004743             0.012441
3.0               0.004514             0.012114
4.0               0.006074             0.018077
5.0               0.001099             0.000865
6.0               0.003205             0.013183
7.0               0.022951             0.012520

Cluster summary
         SERIOUS_RATE   MEAN_AGE       N
CLUSTER
0.0          0.077423  55.810128  122315
1.0          0.295154  43.483201  101215
```

```
2.0          0.093040  52.766412   79493
3.0          0.236332  48.262638  248134
4.0          0.141540  47.837802   41656
5.0          0.009080  44.396722   85569
6.0          0.061855  48.799073  196541
7.0          0.770589  62.761468   80432
```

11. Export

Save the full dataset with cluster labels appended as `df_clean_with_clusters.csv`. This file is the input for the time series analysis notebook, which examines how cluster membership and seriousness rates evolve over the reporting period.

In [ ]:
```python
# ============================================================
# 11. EXPORT FINAL CLUSTERED DATASET
# ============================================================
df_clean.to_csv(f"{OUTDIR}/df_clean_with_clusters.csv", index=False)
print("✓ Full clustered dataset saved for time series analysis")
```

```
✓ Full clustered dataset saved for time series analysis
```